

Making decisions – and fast – at the onset of a disaster

A key challenge faced by humanitarian agencies during the first hours and days of a disaster response (or even before, in the case of cyclones) is the ability and the pressure to make crucial decisions and answer the following questions:

- How many people will be affected? How many of those people are particularly vulnerable?
- What is the potential scale of the emergency and how big will the response have to be?
- Do we need deployments, field offices, emergency funds?
- What are the priority geographic areas? What are the priority sectors and needs?
- Will we request/release emergency funding? If so, how much?

The main challenge? Making these decisions fast, even though information about the situation is scarce or even nonexistent. There is not enough time to rely on rapid needs assessments and affected people need assistance as soon as possible.

A regional initiative between the Pacific Disaster Centre (PDC), WFP and OCHA is trying to address this data gap using state-of-the-art modeling, secondary data, and close cooperation with responding agencies. The Joint Analysis of Disaster Exposure – JADE - builds on PDC's unique modeling capacity, WFP's expertise in vulnerability, and OCHA's convening role in emergency response and preparedness.

The JADE is a two-step process:

1. Using PDC's All-hazard Impact Model (AIM) the JADE will provide the number of people and infrastructure in affected areas. AIM uses advanced algorithms and superior scientific data sources to estimate the **exposure** of populations, assets and infrastructure to natural hazards. JADE can provide population estimates that are age/sex disaggregated, separated into urban and rural populations, and calculated depending on the time of day. Infrastructure estimates are categorized by sector.
2. The JADE analysis goes beyond exposure, to look at who, among those living in the worst affected areas, was already vulnerable and therefore most likely to be in need of international humanitarian assistance. This is accomplished by using agreed-upon country-specific indicators such as multi-dimensional poverty or food insecurity, to identify those whose coping capacity is most likely to be insufficient in the aftermath of a disaster.

The JADE works even in the most data-poor environment and for a variety of hazard types. It provides three key planning figures.

- The number of people living in ALL areas affected by the disaster. This is useful for Government disaster response agencies as well as those focused on recovery and rehabilitation. Not everyone in this category will require humanitarian assistance. Some people live in areas where infrastructure could have been impacted and may otherwise be indirectly affected.
- The number of people living in the worst affected areas. This is useful for estimating the number of people directly affected by the disaster.
- The number of VULNERABLE people living in the worst affected areas. Given that international humanitarian agencies must prioritize, this figure estimates the top priority for humanitarian agencies and those most likely to be in need of international assistance. These priority figures are disaggregated by age/sex and urban/rural.

These key planning figures are supplemented by estimates on the required number of relief items, and estimates on the total cost of infrastructure, by sector, found in the worst-affected areas.

The JADE provides a key evidence base for informing early decisions in an emergency response. It will be activated in coordination with humanitarian decision-makers on the ground. It can also be used to support contingency planning and on scenarios to provide accurate planning figures to support planning and preparedness actions. This can ensure the international humanitarian community is prepared for the next disaster.

For more information please feel free to contact:

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